EXPERIMENT:1

```
mysql> USE sys;
Database changed
mysql> CREATE TABLE student (
    ->     id INT,
    ->     name VARCHAR(30),
    ->     age INT
    -> );
ERROR 1050 (42S01): Table 'student' already exists
mysql> USE sys;
Database changed
mysql> CREATE TABLE student_details (
    ->     id INT,
    ->     name VARCHAR(30),
    ->     age INT
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> ALTER TABLE student_details
    -> ADD department VARCHAR(30);
Query OK, 0 rows affected (0.09 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> DESC student_details;
+------------+-------------+------+-----+---------+-------+
| Field      | Type        | Null | Key | Default | Extra |
+------------+-------------+------+-----+---------+-------+
| id         | int         | YES  |     | NULL    |       |
| name       | varchar(30) | YES  |     | NULL    |       |
| age        | int         | YES  |     | NULL    |       |
| department | varchar(30) | YES  |     | NULL    |       |
+------------+-------------+------+-----+---------+-------+
4 rows in set (0.01 sec)

mysql> INSERT INTO student_details VALUES
    -> (1, 'Ravi', 20, 'CSE'),
    -> (2, 'Priya', 21, 'IT');
Query OK, 2 rows affected (0.03 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM student_details;
```

```
mysql> INSERT INTO student_details VALUES
    -> (1, 'Ravi', 20, 'CSE'),
    -> (2, 'Priya', 21, 'IT');
Query OK, 2 rows affected (0.03 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM student_details;
+------+-------+------+------------+
| id   | name  | age  | department |
+------+-------+------+------------+
|    1 | Ravi  |   20 | CSE        |
|    2 | Priya |   21 | IT         |
+------+-------+------+------------+
2 rows in set (0.00 sec)

mysql> TRUNCATE TABLE student_details;
Query OK, 0 rows affected (0.06 sec)

mysql> SELECT * FROM student_details;
Empty set (0.00 sec)

mysql> DROP TABLE student_details;
Query OK, 0 rows affected (0.07 sec)
```

```
mysql> SHOW TABLES;
+-----------------------------------------------+
| Tables_in_sys                                 |
+-----------------------------------------------+
| host_summary                                  |
| host_summary_by_file_io                       |
| host_summary_by_file_io_type                  |
| host_summary_by_stages                        |
| host_summary_by_statement_latency             |
| host_summary_by_statement_type                |
| innodb_buffer_stats_by_schema                 |
| innodb_buffer_stats_by_table                  |
| innodb_lock_waits                             |
| io_by_thread_by_latency                       |
| io_global_by_file_by_bytes                    |
| io_global_by_file_by_latency                  |
| io_global_by_wait_by_bytes                    |
| io_global_by_wait_by_latency                  |
| latest_file_io                                |
| memory_by_host_by_current_bytes               |
| memory_by_thread_by_current_bytes             |
| memory_by_user_by_current_bytes               |
| memory_global_by_current_bytes                |
| memory_global_total                           |
| metrics                                       |
| processlist                                   |
| ps_check_lost_instrumentation                 |
| schema_auto_increment_columns                 |
| schema_index_statistics                       |
| schema_object_overview                        |
| schema_redundant_indexes                      |
| schema_table_lock_waits                       |
| schema_table_statistics                       |
| schema_table_statistics_with_buffer           |
| schema_tables_with_full_table_scans           |
| schema_unused_indexes                         |
| session                                       |
| session_ssl_status                            |
| statement_analysis                            |
| statements_with_errors_or_warnings            |
```

```
x$innodb_lock_waits
x$io_by_thread_by_latency
x$io_global_by_file_by_bytes
x$io_global_by_file_by_latency
x$io_global_by_wait_by_bytes
x$io_global_by_wait_by_latency
x$latest_file_io
x$memory_by_host_by_current_bytes
x$memory_by_thread_by_current_bytes
x$memory_by_user_by_current_bytes
x$memory_global_by_current_bytes
x$memory_global_total
x$processlist
x$ps_digest_95th_percentile_by_avg_us
x$ps_digest_avg_latency_distribution
x$ps_schema_table_statistics_io
x$schema_flattened_keys
x$schema_index_statistics
x$schema_table_lock_waits
x$schema_table_statistics
x$schema_table_statistics_with_buffer
x$schema_tables_with_full_table_scans
x$session
x$statement_analysis
x$statements_with_errors_or_warnings
x$statements_with_full_table_scans
x$statements_with_runtimes_in_95th_percentile
x$statements_with_sorting
x$statements_with_temp_tables
x$user_summary
x$user_summary_by_file_io
x$user_summary_by_file_io_type
x$user_summary_by_stages
x$user_summary_by_statement_latency
x$user_summary_by_statement_type
x$wait_classes_global_by_avg_latency
x$wait_classes_global_by_latency
x$waits_by_host_by_latency
x$waits_by_user_by_latency
x$waits_global_by_latency
------------------------------------------
02 rows in set (0.01 sec)
```

EXPERIMENT:2

```
mysql> CREATE TABLE department_details (
    ->     dept_id INT PRIMARY KEY,
    ->     dept_name VARCHAR(30)
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> ^C
mysql> INSERT INTO department_details VALUES
    -> (1, 'CSE'),
    -> (2, 'IT'),
    -> (3, 'ECE');
Query OK, 3 rows affected (0.05 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM department_details;
+---------+-----------+
| dept_id | dept_name |
+---------+-----------+
|       1 | CSE       |
|       2 | IT        |
|       3 | ECE       |
+---------+-----------+
3 rows in set (0.00 sec)

mysql> CREATE TABLE student_constraints (
    ->     student_id INT PRIMARY KEY,
    ->     name VARCHAR(30),
    ->     dept_id INT,
    ->     FOREIGN KEY (dept_id)
    ->     REFERENCES department_details(dept_id)
    -> );
Query OK, 0 rows affected (0.09 sec)

mysql> INSERT INTO student_constraints VALUES
    -> (101, 'Ravi', 1),
    -> (102, 'Priya', 2),
    -> (103, 'Arun', 3);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT * FROM student_constraints;
+------------+-------+---------+
| student_id | name  | dept_id |
+------------+-------+---------+
|        101 | Ravi  |       1 |
|        102 | Priya |       2 |
|        103 | Arun  |       3 |
+------------+-------+---------+
3 rows in set (0.00 sec)

mysql> ALTER TABLE student_constraints
    -> ADD email VARCHAR(50) UNIQUE;
Query OK, 0 rows affected (0.18 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> UPDATE student_constraints
    -> SET email = 'ravi@gmail.com'
    -> WHERE student_id = 101;
Query OK, 1 row affected (0.05 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> CREATE TABLE age_details (
    ->     id INT PRIMARY KEY,
    ->     age INT CHECK (age >= 18)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> ALTER TABLE student_constraints
    -> ADD city VARCHAR(30) DEFAULT 'Chennai';
Query OK, 0 rows affected (0.07 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> INSERT INTO student_constraints
    -> (student_id, name, dept_id, email)
    -> VALUES (104, 'Kumar', 1, 'kumar@gmail.com');
Query OK, 1 row affected (0.05 sec)
```

```
mysql> SELECT * FROM student_constraints;
+------------+-------+---------+-----------------+---------+
| student_id | name  | dept_id | email           | city    |
+------------+-------+---------+-----------------+---------+
|        101 | Ravi  |       1 | ravi@gmail.com  | Chennai |
|        102 | Priya |       2 | NULL            | Chennai |
|        103 | Arun  |       3 | NULL            | Chennai |
|        104 | Kumar |       1 | kumar@gmail.com | Chennai |
+------------+-------+---------+-----------------+---------+
4 rows in set (0.00 sec)

mysql> ALTER TABLE student_constraints
    -> MODIFY name VARCHAR(30) NOT NULL;
Query OK, 0 rows affected (0.24 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> INSERT INTO student_constraints
    -> (student_id, dept_id)
    -> VALUES (105, 2);
ERROR 1364 (HY000): Field 'name' doesn't have a default value
mysql> DESC student_constraints;
+------------+-------------+------+-----+---------+-------+
| Field      | Type        | Null | Key | Default | Extra |
+------------+-------------+------+-----+---------+-------+
| student_id | int         | NO   | PRI | NULL    |       |
| name       | varchar(30) | NO   |     | NULL    |       |
| dept_id    | int         | YES  | MUL | NULL    |       |
| email      | varchar(50) | YES  | UNI | NULL    |       |
| city       | varchar(30) | YES  |     | Chennai |       |
+------------+-------------+------+-----+---------+-------+
5 rows in set (0.00 sec)
```

EXPERIMENT:3

```
mysql> CREATE TABLE dml_student (
    ->     id INT,
    ->     name VARCHAR(30),
    ->     age INT,
    ->     department VARCHAR(30)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO dml_student
    -> VALUES (1, 'Ravi', 20, 'CSE');
Query OK, 1 row affected (0.05 sec)

mysql> INSERT INTO dml_student
    -> VALUES (2, 'Priya', 21, 'IT');
Query OK, 1 row affected (0.05 sec)

mysql> INSERT INTO dml_student
    -> VALUES (3, 'Arun', 19, 'ECE');
Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM dml_student;
+------+-------+------+------------+
| id   | name  | age  | department |
+------+-------+------+------------+
|    1 | Ravi  |   20 | CSE        |
|    2 | Priya |   21 | IT         |
|    3 | Arun  |   19 | ECE        |
+------+-------+------+------------+
3 rows in set (0.00 sec)

mysql> SELECT name, department FROM dml_student;
+-------+------------+
| name  | department |
+-------+------------+
| Ravi  | CSE        |
| Priya | IT         |
| Arun  | ECE        |
+-------+------------+
3 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM dml_student
    -> WHERE department = 'CSE';
+------+------+------+------------+
| id   | name | age  | department |
+------+------+------+------------+
|    1 | Ravi |   20 | CSE        |
+------+------+------+------------+
1 row in set (0.00 sec)

mysql>
```

EXPERIMENT:4

```
mysql> SELECT * FROM dml_student;
+------+-------+------+------------+
| id   | name  | age  | department |
+------+-------+------+------------+
|    1 | Ravi  |   20 | CSE        |
|    2 | Priya |   21 | IT         |
|    3 | Arun  |   19 | ECE        |
+------+-------+------+------------+
3 rows in set (0.00 sec)

mysql> UPDATE dml_student
    -> SET age = 22
    -> WHERE id = 1;
Query OK, 1 row affected (0.05 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT * FROM dml_student;
+------+-------+------+------------+
| id   | name  | age  | department |
+------+-------+------+------------+
|    1 | Ravi  |   22 | CSE        |
|    2 | Priya |   21 | IT         |
|    3 | Arun  |   19 | ECE        |
+------+-------+------+------------+
3 rows in set (0.00 sec)

mysql> ^C
mysql> DELETE FROM dml_student
    -> WHERE id = 3;
Query OK, 1 row affected (0.05 sec)

mysql> SELECT * FROM dml_student;
+------+-------+------+------------+
| id   | name  | age  | department |
+------+-------+------+------------+
|    1 | Ravi  |   22 | CSE        |
|    2 | Priya |   21 | IT         |
+------+-------+------+------------+
2 rows in set (0.00 sec)
```

EXPERIMENT:5

```
mysql>
mysql> CREATE TABLE tcl_student (
    ->     id INT PRIMARY KEY,
    ->     name VARCHAR(30),
    ->     marks INT
    -> ) ENGINE=InnoDB;
Query OK, 0 rows affected (0.08 sec)

mysql> INSERT INTO tcl_student VALUES
    -> (1, 'Ravi', 80),
    -> (2, 'Priya', 85),
    -> (3, 'Arun', 90);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM tcl_student;
+----+-------+-------+
| id | name  | marks |
+----+-------+-------+
|  1 | Ravi  |    80 |
|  2 | Priya |    85 |
|  3 | Arun  |    90 |
+----+-------+-------+
3 rows in set (0.00 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE tcl_student
    -> SET marks = 95
    -> WHERE id = 1;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
```

```
mysql> UPDATE tcl_student
    -> SET marks = 100
    -> WHERE id = 3;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> ROLLBACK TO SAVEPOINT s1;
Query OK, 0 rows affected (0.00 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.05 sec)

mysql> SELECT * FROM tcl_student;
+----+-------+-------+
| id | name  | marks |
+----+-------+-------+
|  1 | Ravi  |    95 |
|  2 | Priya |    90 |
|  3 | Arun  |    90 |
+----+-------+-------+
3 rows in set (0.00 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> DELETE FROM tcl_student
    -> WHERE id = 2;
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> ROLLBACK;
Query OK, 0 rows affected (0.04 sec)

mysql> SELECT * FROM tcl_student;
+----+-------+-------+
| id | name  | marks |
+----+-------+-------+
|  1 | Ravi  |    95 |
|  2 | Priya |    90 |
|  3 | Arun  |    90 |
+----+-------+-------+
3 rows in set (0.00 sec)
```

EXPERIMENT:6

```
mysql> CREATE USER 'student_user'@'localhost'
    -> IDENTIFIED BY 'student123';
Query OK, 0 rows affected (0.01 sec)

mysql> GRANT SELECT, INSERT
    -> ON sys.*
    -> TO 'student_user'@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql> SHOW GRANTS FOR 'student_user'@'localhost';
+-----------------------------------------------------------------+
| Grants for student_user@localhost                               |
+-----------------------------------------------------------------+
| GRANT USAGE ON *.* TO `student_user`@`localhost`                |
| GRANT SELECT, INSERT ON `sys`.* TO `student_user`@`localhost` |
+-----------------------------------------------------------------+
2 rows in set (0.00 sec)

mysql> REVOKE INSERT
    -> ON sys.*
    -> FROM 'student_user'@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql> SHOW GRANTS FOR 'student_user'@'localhost';
+-------------------------------------------------------+
| Grants for student_user@localhost                     |
+-------------------------------------------------------+
| GRANT USAGE ON *.* TO `student_user`@`localhost`      |
| GRANT SELECT ON `sys`.* TO `student_user`@`localhost` |
+-------------------------------------------------------+
2 rows in set (0.00 sec)
```

EXPERIMENT:7

```
mysql> CREATE TABLE select_student (
    ->     id INT,
    ->     name VARCHAR(30),
    ->     age INT,
    ->     department VARCHAR(30)
    -> );
Query OK, 0 rows affected (0.05 sec)

mysql> INSERT INTO select_student VALUES
    -> (1, 'Ravi', 20, 'CSE'),
    -> (2, 'Priya', 21, 'IT'),
    -> (3, 'Arun', 19, 'ECE'),
    -> (4, 'Rahul', 22, 'CSE'),
    -> (5, 'Anu', 20, 'IT');
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM select_student;
+------+-------+------+------------+
| id   | name  | age  | department |
+------+-------+------+------------+
|    1 | Ravi  |   20 | CSE        |
|    2 | Priya |   21 | IT         |
|    3 | Arun  |   19 | ECE        |
|    4 | Rahul |   22 | CSE        |
|    5 | Anu   |   20 | IT         |
+------+-------+------+------------+
5 rows in set (0.00 sec)

mysql> SELECT * FROM select_student
    -> WHERE department = 'CSE';
+------+-------+------+------------+
| id   | name  | age  | department |
+------+-------+------+------------+
|    1 | Ravi  |   20 | CSE        |
|    4 | Rahul |   22 | CSE        |
+------+-------+------+------------+
```

```
mysql> SELECT * FROM select_student
    -> WHERE age > 20;
+------+-------+------+------------+
| id   | name  | age  | department |
+------+-------+------+------------+
|    2 | Priya |   21 | IT         |
|    4 | Rahul |   22 | CSE        |
+------+-------+------+------------+
2 rows in set (0.00 sec)

mysql> +------+------+------+------------+
    -> | id   | name | age  | department |
    -> +------+------+------+------------+
    -> | 2    | Priya| 21   | IT         |
    -> | 4    | Rahul| 22   | CSE        |
    -> +------+------+------+------------+
    -> 2 rows in set
    -> SELECT * FROM select_student
    -> WHERE name LIKE 'R%';
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual t
hat corresponds to your MySQL server version for the right syntax to use nea
r '+------+------+------+------------+
| id   | name | age  | department |
+------+' at line 1
mysql> SELECT * FROM select_student
    -> WHERE name LIKE 'R%';
+------+-------+------+------------+
| id   | name  | age  | department |
+------+-------+------+------------+
|    1 | Ravi  |   20 | CSE        |
|    4 | Rahul |   22 | CSE        |
+------+-------+------+------------+
2 rows in set (0.01 sec)

mysql> SELECT * FROM select_student
    -> WHERE name LIKE '%i';
+------+------+------+------------+
| id   | name | age  | department |
+------+------+------+------------+
|    1 | Ravi |   20 | CSE        |
```

```
mysql> SELECT * FROM select_student
    -> WHERE name LIKE '%u%';
+------+-------+------+------------+
| id   | name  | age  | department |
+------+-------+------+------------+
|    3 | Arun  |   19 | ECE        |
|    4 | Rahul |   22 | CSE        |
|    5 | Anu   |   20 | IT         |
+------+-------+------+------------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM select_student
    -> WHERE name LIKE '___';
+------+------+------+------------+
| id   | name | age  | department |
+------+------+------+------------+
|    5 | Anu  |   20 | IT         |
+------+------+------+------------+
1 row in set (0.00 sec)
```

EXPERIMENT:8

```
mysql> CREATE TABLE aggregate_student (
    ->     id INT,
    ->     name VARCHAR(30),
    ->     age INT,
    ->     marks INT,
    ->     department VARCHAR(30)
    -> );
Query OK, 0 rows affected (0.05 sec)

mysql> INSERT INTO aggregate_student VALUES
    -> (1, 'Ravi', 20, 85, 'CSE'),
    -> (2, 'Priya', 21, 90, 'IT'),
    -> (3, 'Arun', 19, 75, 'ECE'),
    -> (4, 'Rahul', 22, 95, 'CSE'),
    -> (5, 'Anu', 20, 80, 'IT');
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM aggregate_student;
+------+-------+------+-------+------------+
| id   | name  | age  | marks | department |
+------+-------+------+-------+------------+
|    1 | Ravi  |   20 |    85 | CSE        |
|    2 | Priya |   21 |    90 | IT         |
|    3 | Arun  |   19 |    75 | ECE        |
|    4 | Rahul |   22 |    95 | CSE        |
|    5 | Anu   |   20 |    80 | IT         |
+------+-------+------+-------+------------+
5 rows in set (0.00 sec)

mysql> SELECT * FROM aggregate_student
    -> WHERE marks BETWEEN 80 AND 90;
+------+-------+------+-------+------------+
| id   | name  | age  | marks | department |
+------+-------+------+-------+------------+
|    1 | Ravi  |   20 |    85 | CSE        |
|    2 | Priya |   21 |    90 | IT         |
|    5 | Anu   |   20 |    80 | IT         |
+------+-------+------+-------+------------+
```

```
mysql> SELECT COUNT(*) FROM aggregate_student
    -> WHERE department IN ('CSE','IT');
+----------+
| COUNT(*) |
+----------+
|        4 |
+----------+
1 row in set (0.01 sec)

mysql> SELECT SUM(marks) FROM aggregate_student;
+------------+
| SUM(marks) |
+------------+
|        425 |
+------------+
1 row in set (0.00 sec)

mysql> SELECT AVG(marks) FROM aggregate_student;
+------------+
| AVG(marks) |
+------------+
|    85.0000 |
+------------+
1 row in set (0.00 sec)

mysql> SELECT MAX(marks) FROM aggregate_student;
+------------+
| MAX(marks) |
+------------+
|         95 |
+------------+
1 row in set (0.00 sec)

mysql> SELECT MIN(marks) FROM aggregate_student;
+------------+
| MIN(marks) |
+------------+
|         75 |
+------------+
1 row in set (0.00 sec)
```

EXPERIMENT:9

```
mysql> CREATE TABLE group_student (
    ->     id INT,
    ->     name VARCHAR(30),
    ->     department VARCHAR(30),
    ->     marks INT
    -> );
Query OK, 0 rows affected (0.05 sec)

mysql> INSERT INTO group_student VALUES
    -> (1, 'Ravi', 'CSE', 85),
    -> (2, 'Priya', 'IT', 90),
    -> (3, 'Arun', 'CSE', 75),
    -> (4, 'Rahul', 'IT', 95),
    -> (5, 'Anu', 'ECE', 80),
    -> (6, 'Kumar', 'CSE', 88);
Query OK, 6 rows affected (0.02 sec)
Records: 6  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM group_student;
+------+-------+------------+-------+
| id   | name  | department | marks |
+------+-------+------------+-------+
|    1 | Ravi  | CSE        |    85 |
|    2 | Priya | IT         |    90 |
|    3 | Arun  | CSE        |    75 |
|    4 | Rahul | IT         |    95 |
|    5 | Anu   | ECE        |    80 |
|    6 | Kumar | CSE        |    88 |
+------+-------+------------+-------+
6 rows in set (0.00 sec)

mysql> SELECT department, COUNT(*)
    -> FROM group_student
    -> GROUP BY department;
+------------+----------+
| department | COUNT(*) |
+------------+----------+
```

```
+------------+----------+
| department | COUNT(*) |
+------------+----------+
| CSE        |        3 |
| IT         |        2 |
+------------+----------+
2 rows in set (0.00 sec)

mysql> SELECT * FROM group_student
    -> ORDER BY marks ASC;
+------+-------+------------+-------+
| id   | name  | department | marks |
+------+-------+------------+-------+
|    3 | Arun  | CSE        |    75 |
|    5 | Anu   | ECE        |    80 |
|    1 | Ravi  | CSE        |    85 |
|    6 | Kumar | CSE        |    88 |
|    2 | Priya | IT         |    90 |
|    4 | Rahul | IT         |    95 |
+------+-------+------------+-------+
6 rows in set (0.00 sec)

mysql> SELECT * FROM group_student
    -> ORDER BY marks DESC;
+------+-------+------------+-------+
| id   | name  | department | marks |
+------+-------+------------+-------+
|    4 | Rahul | IT         |    95 |
|    2 | Priya | IT         |    90 |
|    6 | Kumar | CSE        |    88 |
|    1 | Ravi  | CSE        |    85 |
|    5 | Anu   | ECE        |    80 |
|    3 | Arun  | CSE        |    75 |
+------+-------+------------+-------+
6 rows in set (0.00 sec)
```

EXPERIMENT:10

```
mysql> CREATE TABLE query_student (
    ->     id INT,
    ->     name VARCHAR(30),
    ->     department VARCHAR(30),
    ->     marks INT
    -> );
Query OK, 0 rows affected (0.05 sec)

mysql> INSERT INTO query_student VALUES
    -> (1, 'Ravi', 'CSE', 85),
    -> (2, 'Priya', 'IT', 90),
    -> (3, 'Arun', 'CSE', 75),
    -> (4, 'Rahul', 'IT', 95),
    -> (5, 'Anu', 'ECE', 80);
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM query_student;
+------+-------+------------+-------+
| id   | name  | department | marks |
+------+-------+------------+-------+
|    1 | Ravi  | CSE        |    85 |
|    2 | Priya | IT         |    90 |
|    3 | Arun  | CSE        |    75 |
|    4 | Rahul | IT         |    95 |
|    5 | Anu   | ECE        |    80 |
+------+-------+------------+-------+
5 rows in set (0.00 sec)

mysql> SELECT * FROM query_student
    -> WHERE marks > (
    ->     SELECT AVG(marks)
    ->     FROM query_student
    -> );
```

```
mysql> SELECT * FROM query_student
    -> WHERE marks > (
    ->     SELECT AVG(marks)
    ->     FROM query_student
    -> );
+------+-------+------------+-------+
| id   | name  | department | marks |
+------+-------+------------+-------+
|    2 | Priya | IT         |    90 |
|    4 | Rahul | IT         |    95 |
+------+-------+------------+-------+
2 rows in set (0.00 sec)

mysql> SELECT * FROM query_student s
    -> WHERE marks > (
    ->     SELECT AVG(marks)
    ->     FROM query_student
    ->     WHERE department = s.department
    -> );
+------+-------+------------+-------+
| id   | name  | department | marks |
+------+-------+------------+-------+
|    1 | Ravi  | CSE        |    85 |
|    4 | Rahul | IT         |    95 |
+------+-------+------------+-------+
2 rows in set (0.00 sec)
```

EXPERIMENT:11

```
mysql> CREATE TABLE join_department (
    ->     dept_id INT,
    ->     dept_name VARCHAR(30)
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> INSERT INTO join_department VALUES
    -> (1, 'CSE'),
    -> (2, 'IT'),
    -> (3, 'ECE');
Query OK, 3 rows affected (0.02 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> CREATE TABLE join_student (
    ->     id INT,
    ->     name VARCHAR(30),
    ->     dept_id INT
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO join_student VALUES
    -> (101, 'Ravi', 1),
    -> (102, 'Priya', 2),
    -> (103, 'Arun', 1),
    -> (104, 'Rahul', 4);
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM join_student;
+------+-------+---------+
| id   | name  | dept_id |
+------+-------+---------+
|  101 | Ravi  |       1 |
|  102 | Priya |       2 |
|  103 | Arun  |       1 |
|  104 | Rahul |       4 |
+------+-------+---------+
4 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM join_department;
+---------+-----------+
| dept_id | dept_name |
+---------+-----------+
|       1 | CSE       |
|       2 | IT        |
|       3 | ECE       |
+---------+-----------+
3 rows in set (0.00 sec)

mysql>
mysql> SELECT join_student.id,
    ->        join_student.name,
    ->        join_department.dept_name
    -> FROM join_student, join_department
    -> WHERE join_student.dept_id = join_department.dept_id;
+------+-------+-----------+
| id   | name  | dept_name |
+------+-------+-----------+
|  101 | Ravi  | CSE       |
|  102 | Priya | IT        |
|  103 | Arun  | CSE       |
+------+-------+-----------+
3 rows in set (0.00 sec)

mysql> SELECT s.id, s.name, d.dept_name
    -> FROM join_student s
    -> INNER JOIN join_department d
    -> ON s.dept_id = d.dept_id;
+------+-------+-----------+
| id   | name  | dept_name |
+------+-------+-----------+
|  101 | Ravi  | CSE       |
|  102 | Priya | IT        |
|  103 | Arun  | CSE       |
+------+-------+-----------+
3 rows in set (0.00 sec)
```

```
mysql> SELECT s.id, s.name, d.dept_name
    -> FROM join_student s
    -> LEFT OUTER JOIN join_department d
    -> ON s.dept_id = d.dept_id;
+------+-------+-----------+
| id   | name  | dept_name |
+------+-------+-----------+
|  101 | Ravi  | CSE       |
|  102 | Priya | IT        |
|  103 | Arun  | CSE       |
|  104 | Rahul | NULL      |
+------+-------+-----------+
4 rows in set (0.00 sec)
```

EXPERIMENT:12

```
mysql> SELECT * FROM view_index_student;
+----+-------+------------+-------+
| id | name  | department | marks |
+----+-------+------------+-------+
|  1 | Ravi  | CSE        |    85 |
|  2 | Priya | IT         |    90 |
|  3 | Arun  | CSE        |    75 |
|  4 | Rahul | ECE        |    80 |
+----+-------+------------+-------+
4 rows in set (0.00 sec)

mysql> CREATE VIEW student_view AS
    -> SELECT id, name, department
    -> FROM view_index_student;
Query OK, 0 rows affected (0.02 sec)

mysql> SELECT * FROM student_view;
+----+-------+------------+
| id | name  | department |
+----+-------+------------+
|  1 | Ravi  | CSE        |
|  2 | Priya | IT         |
|  3 | Arun  | CSE        |
|  4 | Rahul | ECE        |
+----+-------+------------+
```

```
mysql> CREATE INDEX idx_student_name
    -> ON view_index_student(name);
Query OK, 0 rows affected (0.06 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM view_index_student
    -> WHERE name = 'Ravi';
+----+------+------------+-------+
| id | name | department | marks |
+----+------+------------+-------+
|  1 | Ravi | CSE        |    85 |
+----+------+------------+-------+
1 row in set (0.00 sec)

mysql> SHOW INDEX FROM view_index_student;
+--------------------+------------+------------------+--------------+-
-----------+----------+------------+----------+--------+------+----
-------+---------+---------------+---------+------------+
| Table              | Non_unique | Key_name         | Seq_in_index |
Column_name | Collation | Cardinality | Sub_part | Packed | Null | Ind
ex_type | Comment | Index_comment | Visible | Expression |
+--------------------+------------+------------------+--------------+-
-----------+----------+------------+----------+--------+------+----
-------+---------+---------------+---------+------------+
| view_index_student |          0 | PRIMARY          |            1 |
id          | A         |           4 |     NULL |   NULL |      | BTR
EE      |         |               | YES     | NULL       |
| view_index_student |          1 | idx_student_name |            1 |
name        | A         |           4 |     NULL |   NULL | YES  | BTR
EE      |         |               | YES     | NULL       |
+--------------------+------------+------------------+--------------+-
-----------+----------+------------+----------+--------+------+----
-------+---------+---------------+---------+------------+
2 rows in set (0.02 sec)
```

```
mysql> DROP VIEW student_view;
Query OK, 0 rows affected (0.02 sec)

mysql> DROP INDEX idx_student_name
    -> ON view_index_student;
Query OK, 0 rows affected (0.03 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

EXPERIMENT:13

```
mysql> SELECT * FROM view_index_student;
+----+-------+------------+-------+
| id | name  | department | marks |
+----+-------+------------+-------+
|  1 | Ravi  | CSE        |    85 |
|  2 | Priya | IT         |    90 |
|  3 | Arun  | CSE        |    75 |
|  4 | Rahul | ECE        |    80 |
+----+-------+------------+-------+
4 rows in set (0.00 sec)

mysql> CREATE VIEW student_view AS
    -> SELECT id, name, department
    -> FROM view_index_student;
Query OK, 0 rows affected (0.02 sec)

mysql> SELECT * FROM student_view;
+----+-------+------------+
| id | name  | department |
+----+-------+------------+
|  1 | Ravi  | CSE        |
|  2 | Priya | IT         |
|  3 | Arun  | CSE        |
|  4 | Rahul | ECE        |
+----+-------+------------+
4 rows in set (0.01 sec)

mysql> CREATE INDEX idx_student_name
    -> ON view_index_student(name);
Query OK, 0 rows affected (0.06 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM view_index_student
    -> WHERE name = 'Ravi';
```

```
mysql> SELECT * FROM view_index_student
    -> WHERE name = 'Ravi';
+----+------+------------+-------+
| id | name | department | marks |
+----+------+------------+-------+
|  1 | Ravi | CSE        |    85 |
+----+------+------------+-------+
1 row in set (0.00 sec)

mysql> SHOW INDEX FROM view_index_student;
+--------------------+------------+-----------------+--------------+-
-----------+-----------+-------------+----------+--------+------+----
--------+---------+---------------+---------+------------+
| Table              | Non_unique | Key_name        | Seq_in_index |
Column_name | Collation | Cardinality | Sub_part | Packed | Null | Ind
ex_type | Comment | Index_comment | Visible | Expression |
+--------------------+------------+-----------------+--------------+-
-----------+-----------+-------------+----------+--------+------+----
--------+---------+---------------+---------+------------+
| view_index_student |          0 | PRIMARY         |            1 |
id          | A         |           4 |     NULL |   NULL |      | BTR
EE      |         |               | YES     | NULL       |
| view_index_student |          1 | idx_student_name |           1 |
name        | A         |           4 |     NULL |   NULL | YES  | BTR
EE      |         |               | YES     | NULL       |
+--------------------+------------+-----------------+--------------+-
-----------+-----------+-------------+----------+--------+------+----
--------+---------+---------------+---------+------------+
2 rows in set (0.02 sec)

mysql> DROP VIEW student_view;
Query OK, 0 rows affected (0.02 sec)

mysql> DROP INDEX idx_student_name
    -> ON view_index_student;
Query OK, 0 rows affected (0.03 sec)
```

```
Query OK, 0 rows affected (0.03 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> CREATE TABLE auto_student (
    ->     id INT AUTO_INCREMENT PRIMARY KEY,
    ->     name VARCHAR(30),
    ->     department VARCHAR(20)
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> INSERT INTO auto_student (name, department)
    -> VALUES ('Ravi', 'CSE');
Query OK, 1 row affected (0.01 sec)

mysql>
mysql> INSERT INTO auto_student (name, department)
    -> VALUES ('Priya', 'IT');
Query OK, 1 row affected (0.01 sec)

mysql>
mysql> INSERT INTO auto_student (name, department)
    -> VALUES ('Arun', 'ECE');
Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM auto_student;
+----+-------+------------+
| id | name  | department |
+----+-------+------------+
|  1 | Ravi  | CSE        |
|  2 | Priya | IT         |
|  3 | Arun  | ECE        |
+----+-------+------------+
3 rows in set (0.00 sec)

mysql> INSERT INTO auto_student (name, department)
    -> VALUES ('Rahul', 'CSE');
```

```
mysql> INSERT INTO auto_student (name, department)
    -> VALUES ('Rahul', 'CSE');
Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM auto_student;
+----+-------+------------+
| id | name  | department |
+----+-------+------------+
|  1 | Ravi  | CSE        |
|  2 | Priya | IT         |
|  3 | Arun  | ECE        |
|  4 | Rahul | CSE        |
+----+-------+------------+
4 rows in set (0.00 sec)

mysql> SHOW TABLE STATUS LIKE 'auto_student';
+--------------+--------+---------+------------+------+--------------
-+-------------+----------------+-------------+-----------+---------
-------+--------------------+--------------------+------------+-----
--------------+----------+----------------+---------+
| Name         | Engine | Version | Row_format | Rows | Avg_row_length
 | Data_length | Max_data_length | Index_length | Data_free | Auto_inc
rement | Create_time         | Update_time         | Check_time | Coll
ation           | Checksum | Create_options | Comment |
+--------------+--------+---------+------------+------+--------------
-+-------------+----------------+-------------+-----------+---------
-------+--------------------+--------------------+------------+-----
--------------+----------+----------------+---------+
| auto_student | InnoDB |      10 | Dynamic    |    4 |           4096
 |       16384 |               0 |           0 |         0 |
     5 | 2026-10-30 22:26:24 | 2026-10-30 22:26:43 | NULL       | utf8
mb4_0900_ai_ci |     NULL |                |         |
+--------------+--------+---------+------------+------+--------------
-+-------------+----------------+-------------+-----------+---------
-------+--------------------+--------------------+------------+-----
```

```
mysql> ALTER TABLE auto_student AUTO_INCREMENT = 10;
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> INSERT INTO auto_student (name, department)
    -> VALUES ('Sneha', 'IT');
Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM auto_student;
+----+-------+------------+
| id | name  | department |
+----+-------+------------+
|  1 | Ravi  | CSE        |
|  2 | Priya | IT         |
|  3 | Arun  | ECE        |
|  4 | Rahul | CSE        |
| 10 | Sneha | IT         |
+----+-------+------------+
5 rows in set (0.00 sec)
```

EXPERIMENT:14

```
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE while_example()
    -> BEGIN
    ->     DECLARE i INT DEFAULT 1;
    ->
    ->     WHILE i <= 5 DO
    ->         SELECT i;
    ->         SET i = i + 1;
    ->     END WHILE;
    -> END //
Query OK, 0 rows affected (0.06 sec)

mysql>
mysql> DELIMITER ;
mysql> CALL while_example();
+------+
| i    |
+------+
|    1 |
+------+
1 row in set (0.05 sec)

+------+
| i    |
+------+
|    2 |
+------+
1 row in set (0.05 sec)

+------+
| i    |
+------+
```

```
+------+
| i    |
+------+
|    4 |
+------+
1 row in set (0.05 sec)

+------+
| i    |
+------+
|    5 |
+------+
1 row in set (0.05 sec)

Query OK, 0 rows affected (0.06 sec)

mysql> ^C
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE repeat_example()
    -> BEGIN
    ->     DECLARE i INT DEFAULT 1;
    ->
    ->     REPEAT
    ->         SELECT i;
    ->         SET i = i + 1;
    ->     UNTIL i > 5
    ->     END REPEAT;
    -> END //
Query OK, 0 rows affected (0.05 sec)

mysql>
mysql> DELIMITER ;
```

```
mysql> DELIMITER ;
mysql> CALL repeat_example();
+------+
| i    |
+------+
|    1 |
+------+
1 row in set (0.00 sec)

+------+
| i    |
+------+
|    2 |
+------+
1 row in set (0.00 sec)

+------+
| i    |
+------+
|    3 |
+------+
1 row in set (0.01 sec)

+------+
| i    |
+------+
|    4 |
+------+
1 row in set (0.01 sec)

+------+
| i    |
+------+
|    5 |
+------+
1 row in set (0.01 sec)
```

```
Query OK, 0 rows affected (0.01 sec)

mysql> DROP PROCEDURE while_example;
Query OK, 0 rows affected (0.01 sec)

mysql> DROP PROCEDURE repeat_example;
Query OK, 0 rows affected (0.05 sec)
```

EXPERIMENT:15

```
mysql> CREATE TABLE case_student (
    ->     id INT PRIMARY KEY,
    ->     name VARCHAR(30),
    ->     marks INT
    -> );
Query OK, 0 rows affected (0.08 sec)

mysql> INSERT INTO case_student VALUES
    -> (1, 'Ravi', 95),
    -> (2, 'Priya', 85),
    -> (3, 'Arun', 70),
    -> (4, 'Rahul', 45);
Query OK, 4 rows affected (0.05 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> SELECT name, marks,
    -> CASE
    ->     WHEN marks >= 90 THEN 'A Grade'
    ->     WHEN marks >= 80 THEN 'B Grade'
    ->     WHEN marks >= 60 THEN 'C Grade'
    ->     ELSE 'Fail'
    -> END AS Grade
    -> FROM case_student;
+-------+-------+---------+
| name  | marks | Grade   |
+-------+-------+---------+
| Ravi  |    95 | A Grade |
| Priya |    85 | B Grade |
| Arun  |    70 | C Grade |
| Rahul |    45 | Fail    |
+-------+-------+---------+
4 rows in set (0.00 sec)
```

```
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE loop_example()
    -> BEGIN
    ->     DECLARE i INT DEFAULT 1;
    ->
    ->     myloop: LOOP
    ->         SELECT i;
    ->
    ->         SET i = i + 1;
    ->
    ->         IF i > 5 THEN
    ->             LEAVE myloop;
    ->         END IF;
    ->
    ->     END LOOP myloop;
    -> END //
Query OK, 0 rows affected (0.05 sec)

mysql>
mysql> DELIMITER ;
mysql> CALL loop_example();
+------+
| i    |
+------+
|    1 |
+------+
1 row in set (0.00 sec)

+------+
| i    |
+------+
|    2 |
+------+
```

```
1 row in set (0.00 sec)

+------+
| i    |
+------+
|    2 |
+------+
1 row in set (0.00 sec)

+------+
| i    |
+------+
|    3 |
+------+
1 row in set (0.00 sec)

+------+
| i    |
+------+
|    4 |
+------+
1 row in set (0.01 sec)

+------+
| i    |
+------+
|    5 |
+------+
1 row in set (0.01 sec)

Query OK, 0 rows affected (0.01 sec)

mysql> DROP PROCEDURE loop_example;
Query OK, 0 rows affected (0.04 sec)
```

EXPERIMENT:16

```
mysql> CREATE TABLE procedure_student (
    ->     id INT PRIMARY KEY,
    ->     name VARCHAR(30),
    ->     department VARCHAR(20),
    ->     marks INT
    -> );
Query OK, 0 rows affected (0.05 sec)

mysql> INSERT INTO procedure_student VALUES
    -> (1, 'Ravi', 'CSE', 85),
    -> (2, 'Priya', 'IT', 90),
    -> (3, 'Arun', 'ECE', 75);
Query OK, 3 rows affected (0.05 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE display_students()
    -> BEGIN
    ->     SELECT * FROM procedure_student;
    -> END //
Query OK, 0 rows affected (0.05 sec)

mysql>
mysql> DELIMITER ;
mysql> CALL display_students();
+----+-------+------------+-------+
| id | name  | department | marks |
+----+-------+------------+-------+
|  1 | Ravi  | CSE        |    85 |
|  2 | Priya | IT         |    90 |
|  3 | Arun  | ECE        |    75 |
+----+-------+------------+-------+
3 rows in set (0.00 sec)
```

```
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE search_student(IN student_id INT)
    -> BEGIN
    ->     SELECT * FROM procedure_student
    ->     WHERE id = student_id;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql> CALL search_student(1);
+----+------+------------+-------+
| id | name | department | marks |
+----+------+------------+-------+
|  1 | Ravi | CSE        |    85 |
+----+------+------------+-------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE update_marks(
    ->     IN student_id INT,
    ->     IN new_marks INT
    -> )
    -> BEGIN
    ->     UPDATE procedure_student
    ->     SET marks = new_marks
    ->     WHERE id = student_id;
    -> END //
Query OK, 0 rows affected (0.01 sec)
```

```
mysql>
mysql> DELIMITER ;
mysql> CALL update_marks(1, 95);
Query OK, 1 row affected (0.05 sec)

mysql>
mysql> SELECT * FROM procedure_student;
+----+-------+------------+-------+
| id | name  | department | marks |
+----+-------+------------+-------+
|  1 | Ravi  | CSE        |    95 |
|  2 | Priya | IT         |    90 |
|  3 | Arun  | ECE        |    75 |
+----+-------+------------+-------+
3 rows in set (0.00 sec)

mysql> DROP PROCEDURE display_students;
Query OK, 0 rows affected (0.05 sec)

mysql> DROP PROCEDURE search_student;
Query OK, 0 rows affected (0.05 sec)

mysql> DROP PROCEDURE update_marks;
Query OK, 0 rows affected (0.05 sec)
```

EXPERIMENT:17

```
mysql> CREATE TABLE function_student (
    ->     id INT PRIMARY KEY,
    ->     name VARCHAR(30),
    ->     marks INT
    -> );
Query OK, 0 rows affected (0.08 sec)

mysql> INSERT INTO function_student VALUES
    -> (1, 'Ravi', 85),
    -> (2, 'Priya', 90),
    -> (3, 'Arun', 75);
Query OK, 3 rows affected (0.06 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> DELIMITER //
mysql>
mysql> CREATE FUNCTION calculate_bonus(marks INT)
    -> RETURNS INT
    -> DETERMINISTIC
    -> BEGIN
    ->     RETURN marks + 5;
    -> END //
Query OK, 0 rows affected (0.05 sec)

mysql>
mysql> DELIMITER ;
mysql> SELECT calculate_bonus(85) AS Total_Marks;
+-------------+
| Total_Marks |
+-------------+
|          90 |
+-------------+
1 row in set (0.04 sec)

mysql> SELECT name, marks,
```

```
mysql> SELECT name, marks,
    -> calculate_bonus(marks) AS Updated_Marks
    -> FROM function_student;
+-------+-------+---------------+
| name  | marks | Updated_Marks |
+-------+-------+---------------+
| Ravi  |    85 |            90 |
| Priya |    90 |            95 |
| Arun  |    75 |            80 |
+-------+-------+---------------+
3 rows in set (0.00 sec)

mysql> DROP FUNCTION calculate_bonus;
Query OK, 0 rows affected (0.05 sec)

mysql>
```

EXPERIMENT:18

```
mysql> CREATE TABLE cursor_student (
    ->     id INT PRIMARY KEY,
    ->     name VARCHAR(30),
    ->     marks INT
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO cursor_student VALUES
    -> (1, 'Ravi', 85),
    -> (2, 'Priya', 90),
    -> (3, 'Arun', 75);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM cursor_student;
+----+-------+-------+
| id | name  | marks |
+----+-------+-------+
|  1 | Ravi  |    85 |
|  2 | Priya |    90 |
|  3 | Arun  |    75 |
+----+-------+-------+
3 rows in set (0.00 sec)

mysql> DELIMITER //
```

```
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE cursor_example()
    -> BEGIN
    ->     DECLARE finished INT DEFAULT 0;
    ->     DECLARE student_name VARCHAR(30);
    ->     DECLARE student_marks INT;
    ->
    ->     DECLARE student_cursor CURSOR FOR
    ->         SELECT name, marks FROM cursor_student;
    ->
    ->     DECLARE CONTINUE HANDLER FOR NOT FOUND
    ->         SET finished = 1;
    ->
    ->     OPEN student_cursor;
    ->
    ->     read_loop: LOOP
    ->         FETCH student_cursor
    ->         INTO student_name, student_marks;
    ->
    ->         IF finished = 1 THEN
    ->             LEAVE read_loop;
    ->         END IF;
    ->
    ->         SELECT student_name, student_marks;
    ->     END LOOP;
    ->
    ->     CLOSE student_cursor;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql> CALL cursor_example();
```

```
mysql>
mysql> DELIMITER ;
mysql> CALL cursor_example();
+--------------+---------------+
| student_name | student_marks |
+--------------+---------------+
| Ravi         |            85 |
+--------------+---------------+
1 row in set (0.04 sec)

+--------------+---------------+
| student_name | student_marks |
+--------------+---------------+
| Priya        |            90 |
+--------------+---------------+
1 row in set (0.05 sec)

+--------------+---------------+
| student_name | student_marks |
+--------------+---------------+
| Arun         |            75 |
+--------------+---------------+
1 row in set (0.05 sec)

Query OK, 0 rows affected (0.05 sec)

mysql> DROP PROCEDURE cursor_example;
Query OK, 0 rows affected (0.05 sec)
```

EXPERIMENT:19

```
mysql> CREATE TABLE trigger_student (
    ->     id INT PRIMARY KEY,
    ->     name VARCHAR(30),
    ->     marks INT
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO trigger_student VALUES
    -> (1, 'Ravi', 85),
    -> (2, 'Priya', 90);
Query OK, 2 rows affected (0.01 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql> CREATE TABLE student_audit (
    ->     message VARCHAR(100)
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> DELIMITER //
mysql>
mysql> CREATE TRIGGER student_before_insert
    -> BEFORE INSERT ON trigger_student
    -> FOR EACH ROW
    -> BEGIN
    ->     INSERT INTO student_audit
    ->     VALUES ('New student record is being inserted');
    -> END //
Query OK, 0 rows affected (0.06 sec)

mysql>
mysql> DELIMITER ;
mysql> INSERT INTO trigger_student VALUES
    -> (3, 'Arun', 75);
Query OK, 1 row affected (0.05 sec)
```

```
mysql> SELECT * FROM trigger_student;
+----+-------+-------+
| id | name  | marks |
+----+-------+-------+
|  1 | Ravi  |    85 |
|  2 | Priya |    90 |
|  3 | Arun  |    75 |
+----+-------+-------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM student_audit;
+--------------------------------------+
| message                              |
+--------------------------------------+
| New student record is being inserted |
+--------------------------------------+
1 row in set (0.00 sec)

mysql> DROP TRIGGER student_before_insert;
Query OK, 0 rows affected (0.01 sec)
```

EXPERIMENT:20

```
mysql> SELECT REPLACE('Hello World', 'World', 'MySQL') AS Result;
+-------------+
| Result      |
+-------------+
| Hello MySQL |
+-------------+
1 row in set (0.04 sec)

mysql> SELECT REPEAT('SQL ', 3) AS Result;
+--------------+
| Result       |
+--------------+
| SQL SQL SQL  |
+--------------+
1 row in set (0.04 sec)

mysql> SELECT REVERSE('MYSQL') AS Result;
+--------+
| Result |
+--------+
| LQSYM  |
+--------+
1 row in set (0.04 sec)

mysql> SELECT RIGHT('COMPUTER', 4) AS Result;
+--------+
| Result |
+--------+
| UTER   |
+--------+
```

```
1 row in set (0.04 sec)

mysql> SELECT LEFT('COMPUTER', 4) AS Result;
+--------+
| Result |
+--------+
| COMP   |
+--------+
1 row in set (0.04 sec)

mysql> SELECT RPAD('SQL', 6, '*') AS Result;
+--------+
| Result |
+--------+
| SQL*** |
+--------+
1 row in set (0.04 sec)

mysql> SELECT LPAD('SQL', 6, '*') AS Result;
+--------+
| Result |
+--------+
| ***SQL |
+--------+
1 row in set (0.00 sec)
```